EvoRubric: Self-Evolving Rubric-Driven RL for Open-Ended Generation

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HarnessRSI 63 ★★

Paper

1 What changes, what is kept

1. Shared policy, two roles
Qwen3 learns answers AND rubric writing.
Rubrics: signed requirements / error checks.

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2. Sample responses → propose criteria
8 answers + 4 rubric sets per training query.
Read fixed base rubrics and per-query history.

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3. Frozen checks, not self-judging alone
Initial Qwen3 checks criterion validity;
GPT-OSS-120B grades each answer.

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4. Gate and keep
Nonconstant scores + LOO correlation > 0.2.
Only best eligible group adds after dedup.

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5. Remember, retire, update
Empty pool at start; cap 10; FIFO eviction.
Retire saturated criteria; GRPO updates weights.

All eligible groups reward current answers; only the best enters memory. Consensus is not correctness or a held-out improvement gate.

Reward: base + half history + half current − length penalty. Generator: validity + discrimination + agreement − penalties.

Hybrid system: weights change too; memory is per training query, not a universal inference-time checklist.

2 Higher five-benchmark means



Table 1: unweighted averages, no average SD/CI. RL: 3 training seeds; Base: 3 sampling runs. Margins +3.09 (8B), +2.06 (14B).

Separate domain policies; final judge claude-sonnet-4-6. Held-out HealthBench (500), ResearchQA; also LLMEval-Med, WritingBench, Creative Writing.

4 Where gains are small / absent

- **Uneven gains:** 8B relative gains largest on HealthBench (14.2%) and Creative Writing (8.4%).
- **LLMEval-Med:** 53.75 ± 0.84 vs Static 52.95 ± 0.47 (8B); 59.23 ± 0.91 vs 58.30 ± 0.25 (14B). No significance tests.
- **Human trade-off:** vs GPT-4.1-generator baseline, W/T/L (%) — overall 48/14/38; completeness 36/18/46; factual correctness 42/14/44 (50 Medical prompts).
- **Residual drift:** 8B pool invalidity rises 1.08% → 5.10%; diversity 89.19% → 75.16%.

3 Controls and accounting limits

Medical ablations · Qwen3-8B

Table 2	Health	LLMEval
Full	**45.13**	**53.75**
No Generator loss	41.58	49.93
No discrimination	38.92	51.11
No peer consensus	40.80	50.16
No history pool	42.26	52.41
No Meta-Verifier	43.13	**47.81**

3 matched-run means, no SD/CI. No Generator loss keeps selection rules; no history keeps current criteria. Base LLMEval: 49.25.

Baseline grants: Static: frozen few-shot rubrics. External: frozen same-scale generator. Same data/backbone/policy settings ≠ total-compute match.

Stronger external generator: GPT-4.1 reads responses + existing rubrics: 42.25 / 52.92 vs full 45.13 / 53.75 (3-run means, no SD/CI). Not full DR-Tulu; shared gates/budgets not fully listed.

Validity audit: 8B invalid 5.10% vs 39.06% without verifier; 20 Medical prompts. NOT a leakage audit.

Budget: 4 epochs, 8 × H20. No total Evo cost/GPU-hours or reward-model budget match. GPT-4.1 API only: $185.70/run.

Selection: Numerical defaults given; tuning split, checkpoint rule, baseline trial budgets and overlap audit unreported.

Takeaway

Learning the rubric writer alongside the responder improves the reported means. Independent judging and ablations help; consensus does not prove correctness, and cost parity and leakage protection remain unestablished.

Our read · HarnessRSI

Score 63 · ★★ · recorded level: improver

Reports: held-out tests, seed SDs, ablations, human audits.
Judgment notes: seed variance understated; validity audit is not leakage testing. Score unchanged.